W7 Data Visualisation

Lecture

Agenda

- 7.1 Introduction to Psychology of Data Visualisation Design

Class

Agenda

- Data Visualisation Design

W7.1 Introduction to Psychology of Data Visualisation Design

Lecture

Agenda

- Sensation and perception
- Memory
- Pre-attentive processing
- Colour

References

Johnson (2013) Designing with the Mind in Mind. Chapter 1 and 2

Knaflic (2015) Storytelling with data

Cairo (2012) The Functional Art: An introduction to information graphics and visualization

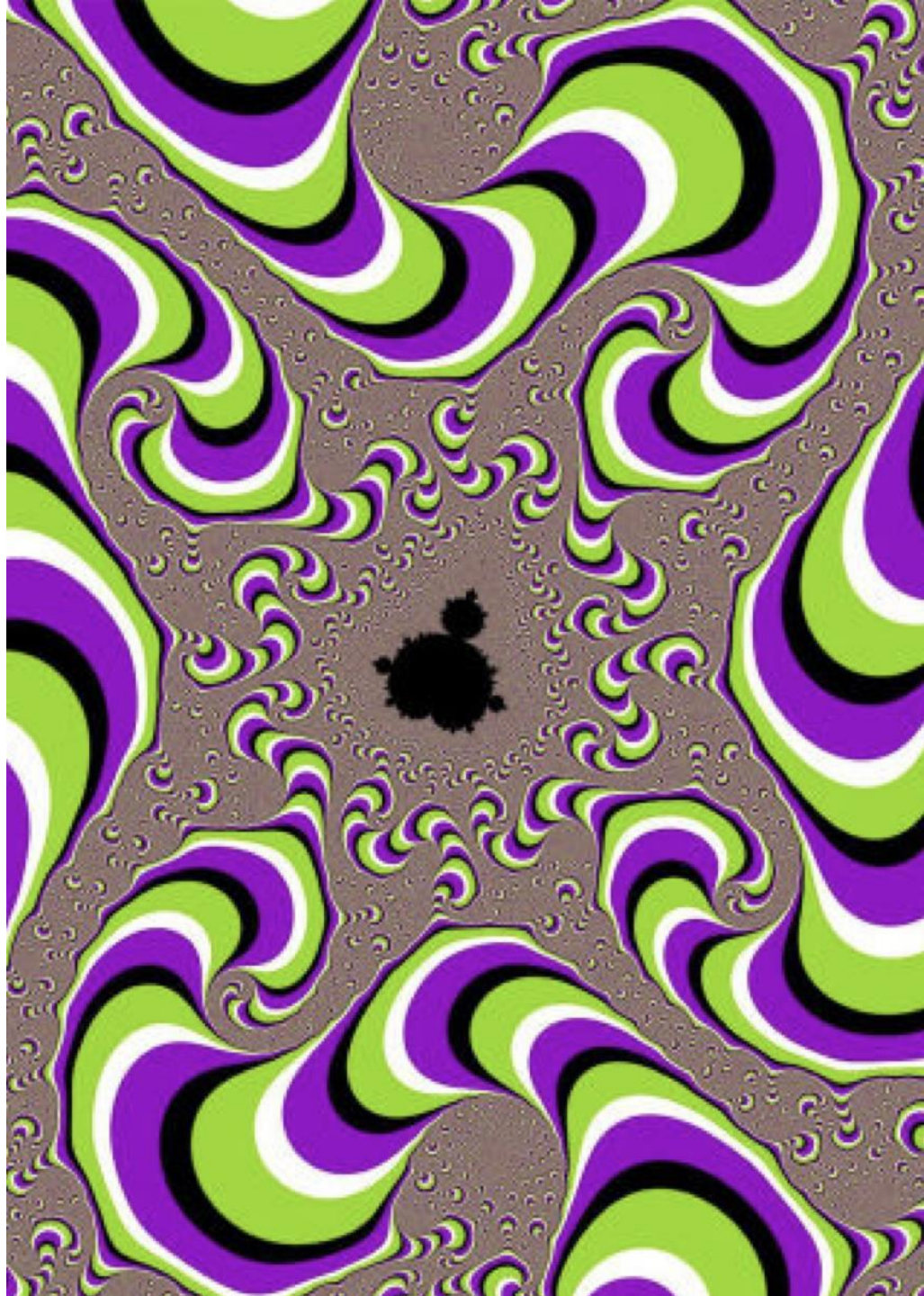
Sensation and Perception

Sensation

- receiving physical stimulation and encoding it into the nervous system

Perception

- the process of interpreting and recognising sensory information, the act of understanding what the sensation was



Perception

Not a true depiction of the world

We see with our brain!

Biased by

- our past experience
- the current context
- our future goals



Red and orange shapes are identical

Size constancy

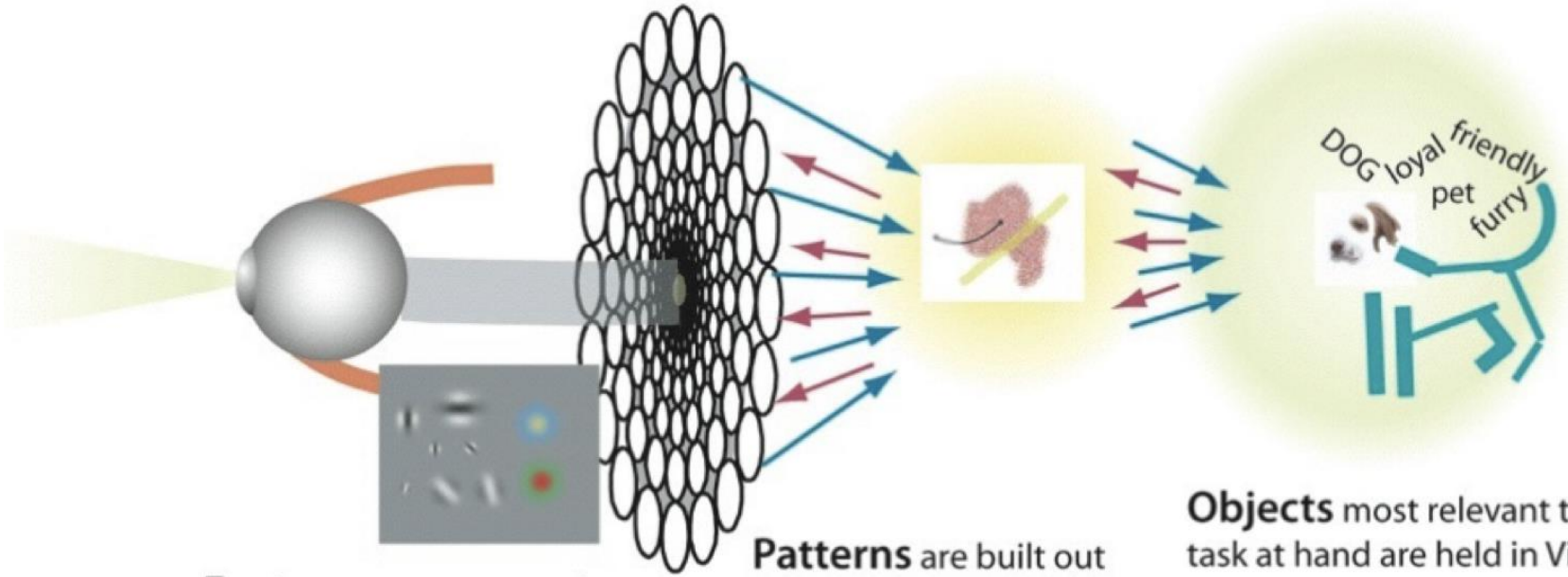


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Size constancy



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Features are processed in parallel from every part of the visual field. Millions of features are processed simultaneously.

Patterns are built out of features depending on attentional demands. Attentional tuning reinforces those most relevant.

Objects most relevant to the task at hand are held in Visual Working Memory. Only between one and three are held at any instant. Objects have both non-visual and visual attributes.

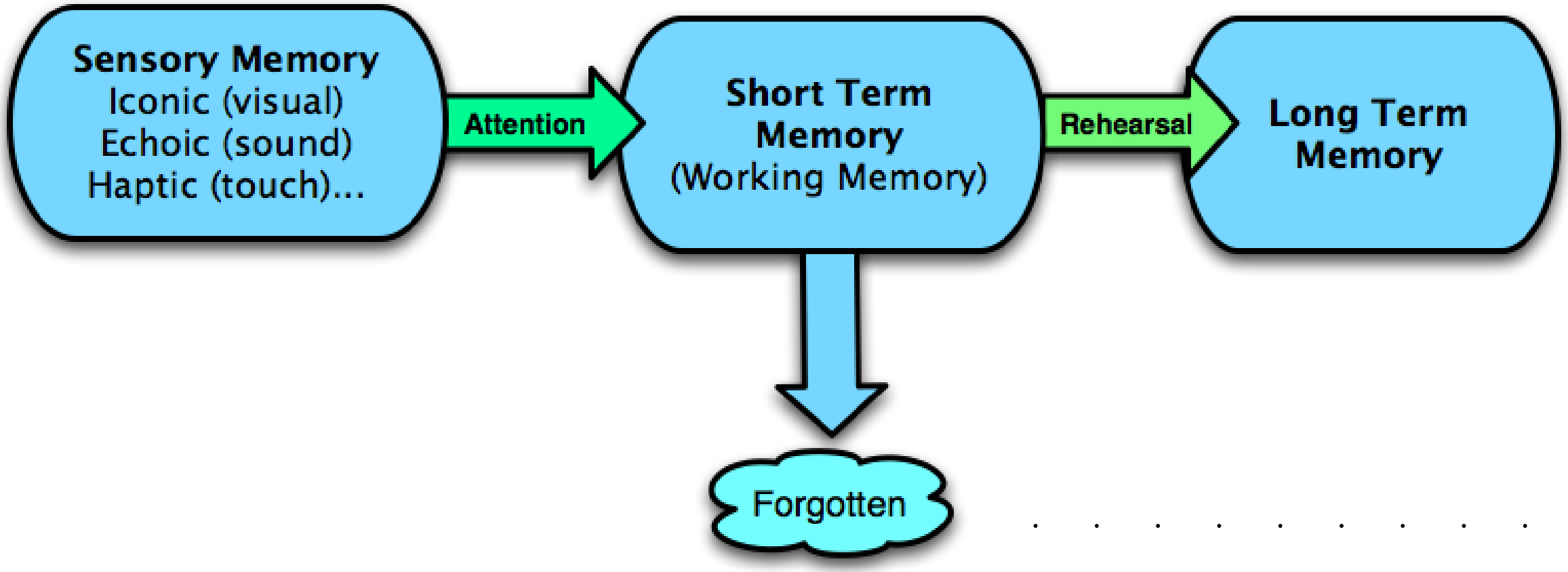
Bottom-up information drives pattern building

Top-down attentional processes reinforce relevant information

Ware (2010)

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Memory



Very simplified version of how we think memory works!!

Iconic Memory

Visual aspect of sensory memory

Ability to quickly detect differences in our environment

Evolutionary history

- detect and escape from predators
- Find food

Attuned to differences in:

- Size
- Colour
- Position
- Motion

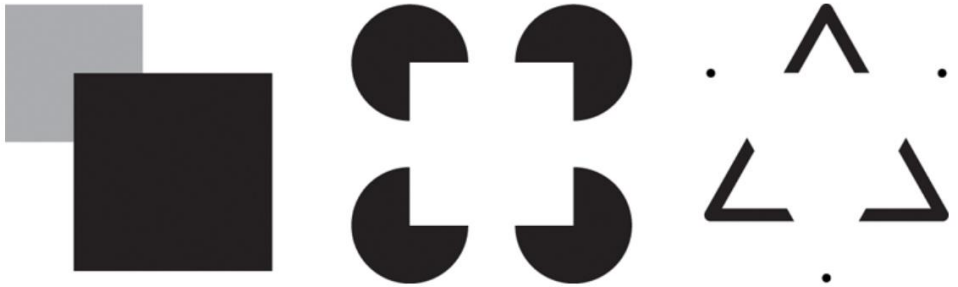


Figure 5.9. Can you see two squares, a white square over four circles, and a white triangle?



Figure 5.11. If you were out walking on the savannah, I am sure you would not stop to figure out what your eyes were seeing before you fe;t the urge to flee. (Photograph by Rich Beckman.)

Cairo (2012)



Ware (2010)

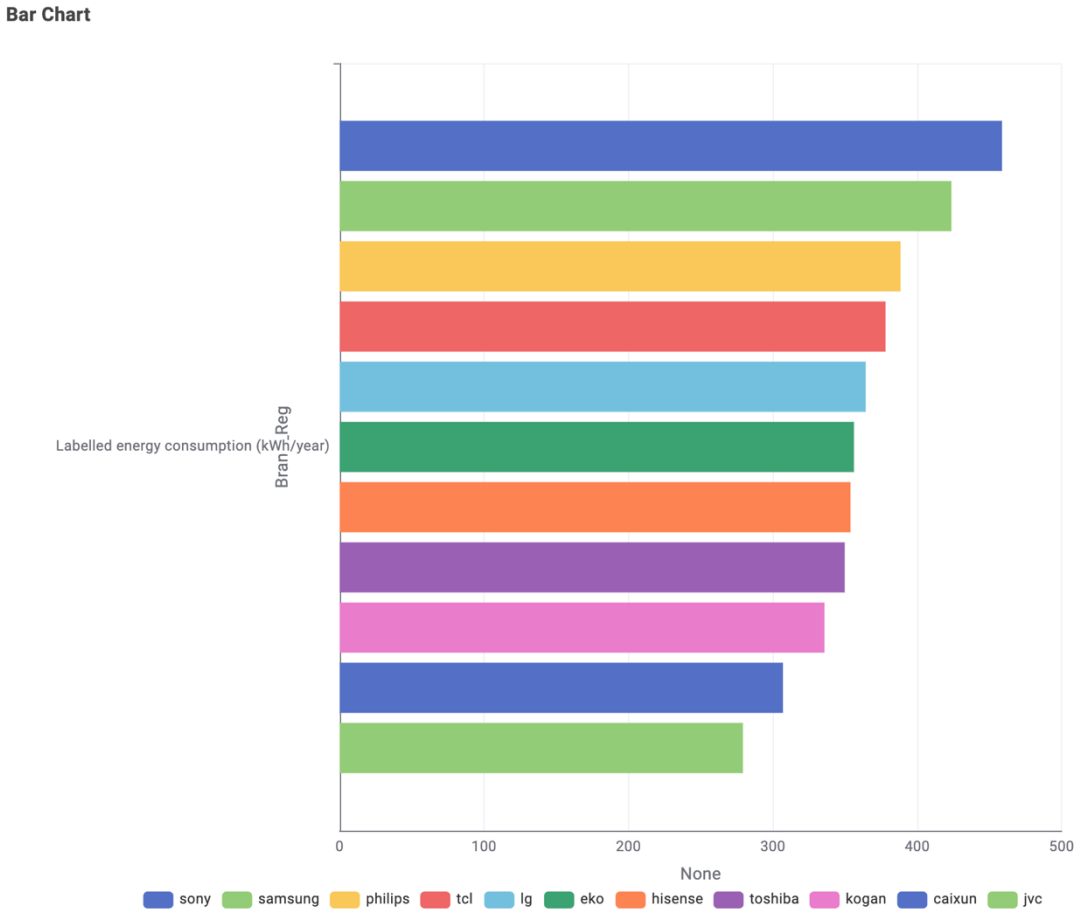
Short term memory

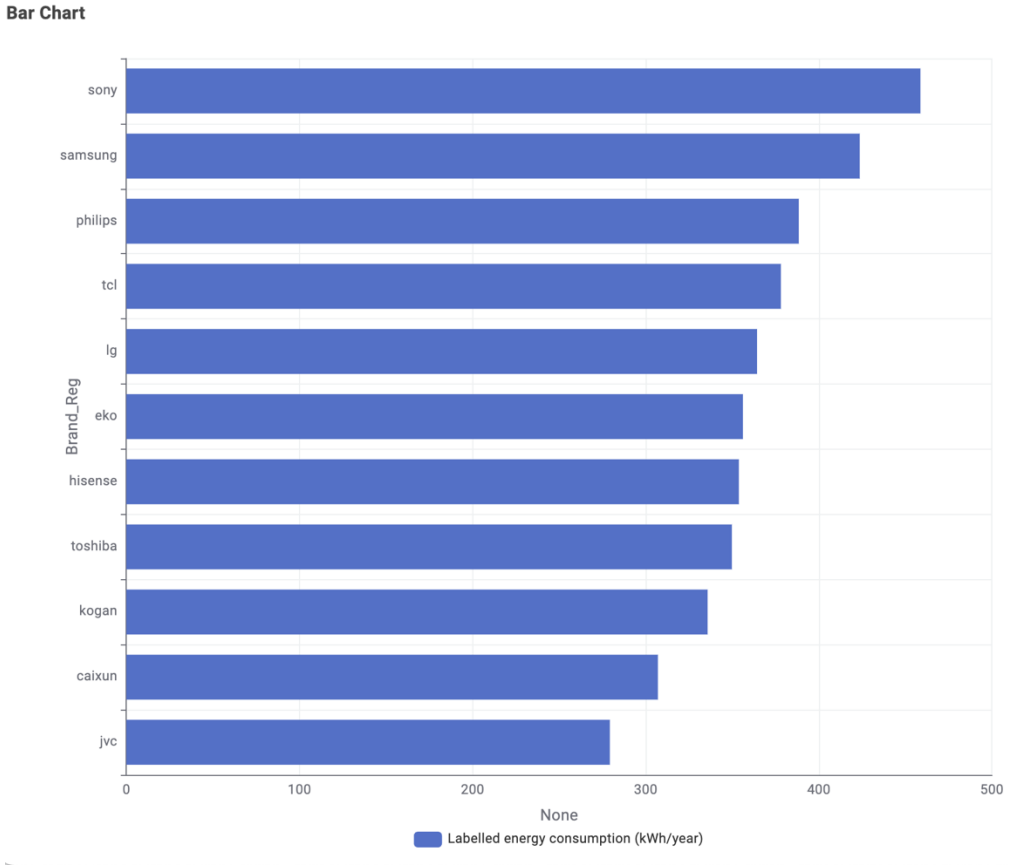
- Limited amount of space
- Around 7 plus or minus 2 (i.e., 5-9) items
- People have difficulty recalling more than 7-9 new bits of information





Lots of categories? Users will not be able to remember which colour goes with which bar… labels on chart better than legend





Don’t make users work too hard to get information!

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Pre-attentive attributes

Quickly count the number of 3s

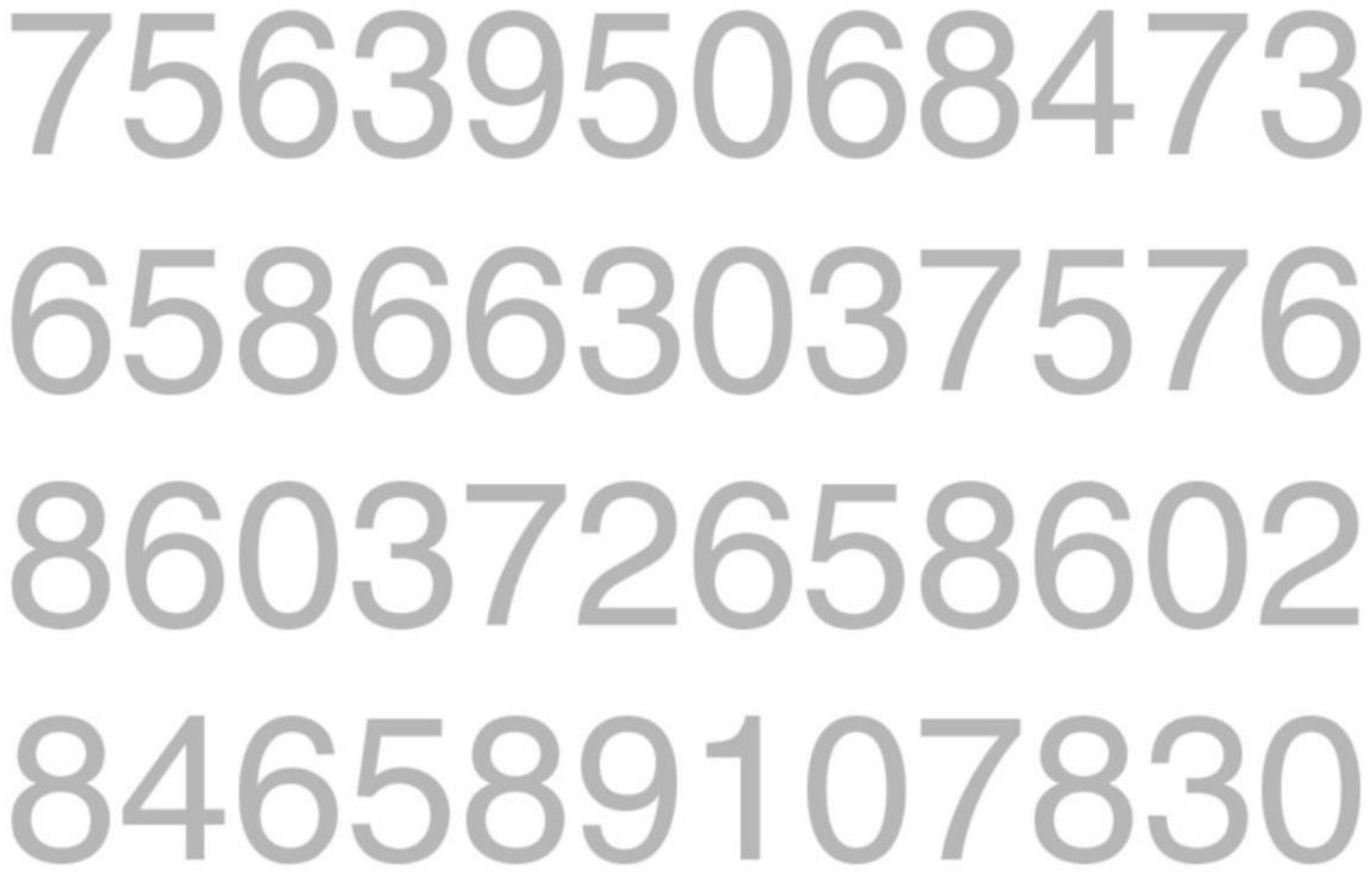


Figure 4.2 Count the 3s example

Knaflic(2015)

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Pre-attentive attributes

Quickly count the number of 3s

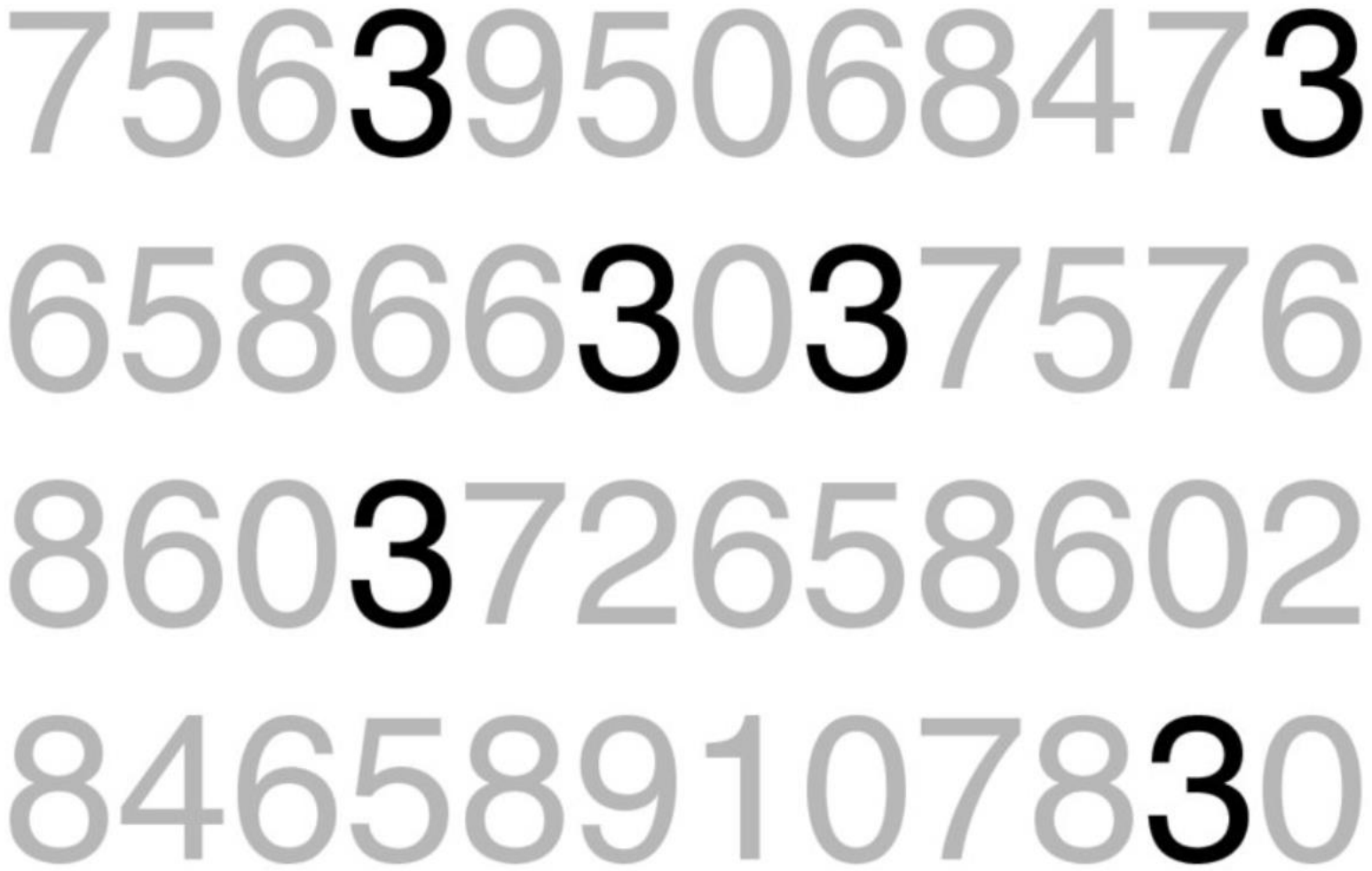


Figure 4.3 Count the 3s example with preattentive attributes

Knaflic(2015)

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Pre-attentive Processing

- Very fast < 200-250 ms
- Targets (i.e., numbers) processed in parallel
- Picks up on contrast between features
- Independent of number of distractors

No need to think about it!

Can use this to draw attention to important features of our charts

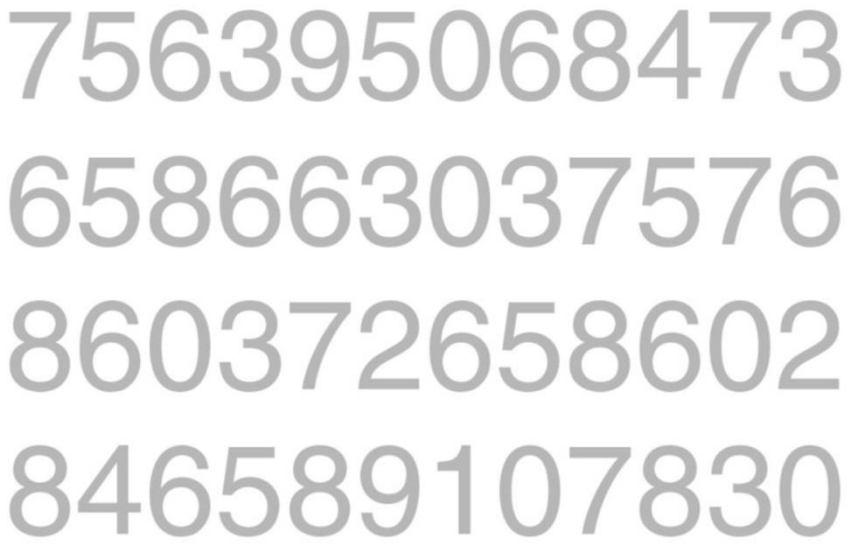


Figure 4.2 Count the 3s example

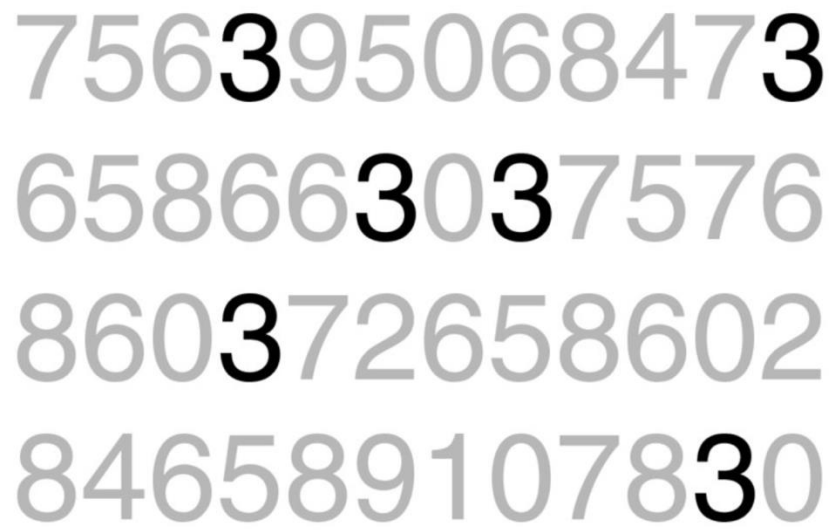


Figure 4.3 Count the 3s example with preattentive attributes

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Pre-attentive Attribu

We are ‘hardwired’ to pick out differences

Can be used to:

- Draw user’s attention to something quickly
- Create visual hierarchy

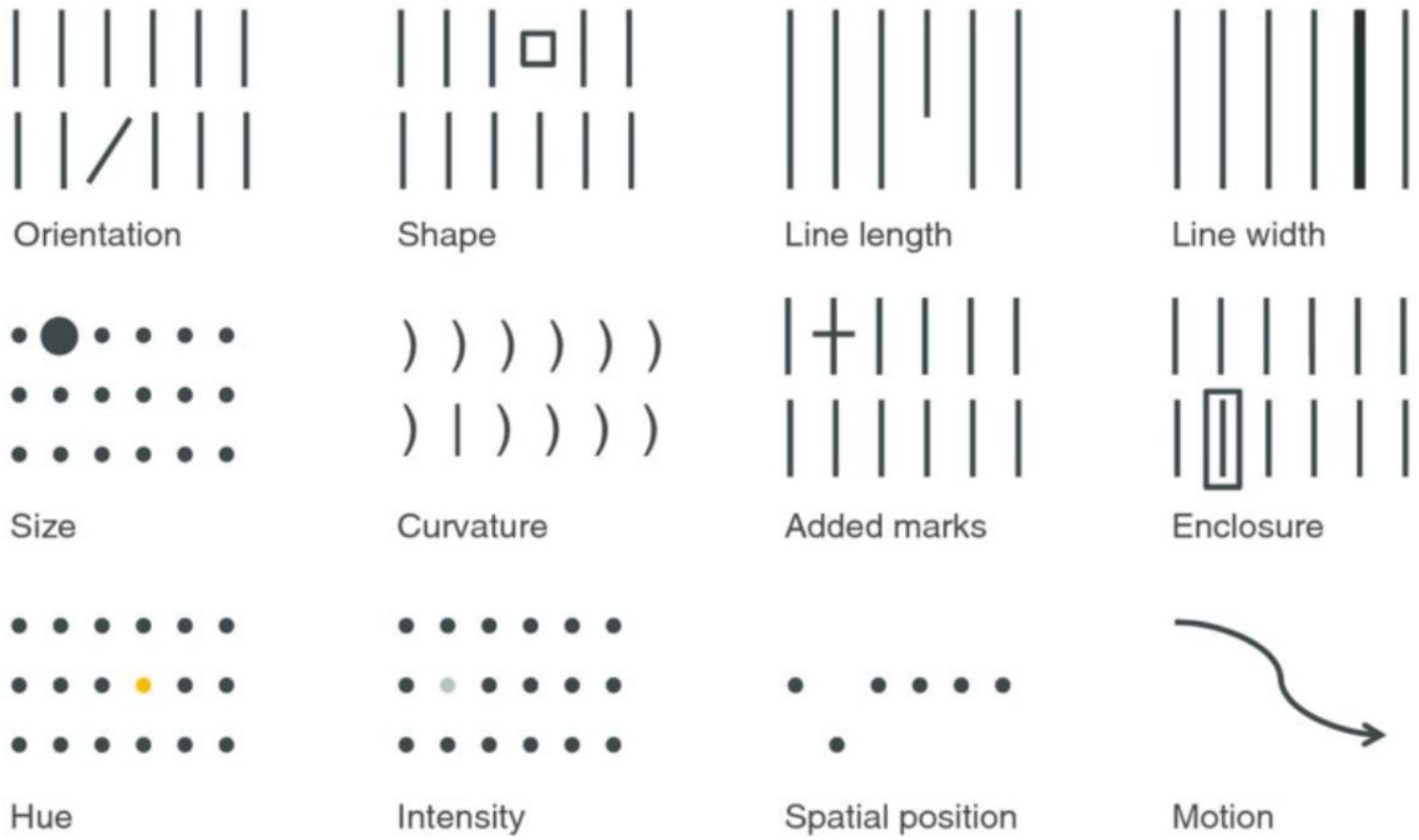


Figure 4.4 Preattentive attributes

Source: Adapted from Stephen Few’s *Show Me the Numbers*, 2004.

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Differences in contrast are easier to see…



Figure 6.1. How quickly can you see the wolf in the trees in each of these illustrations?

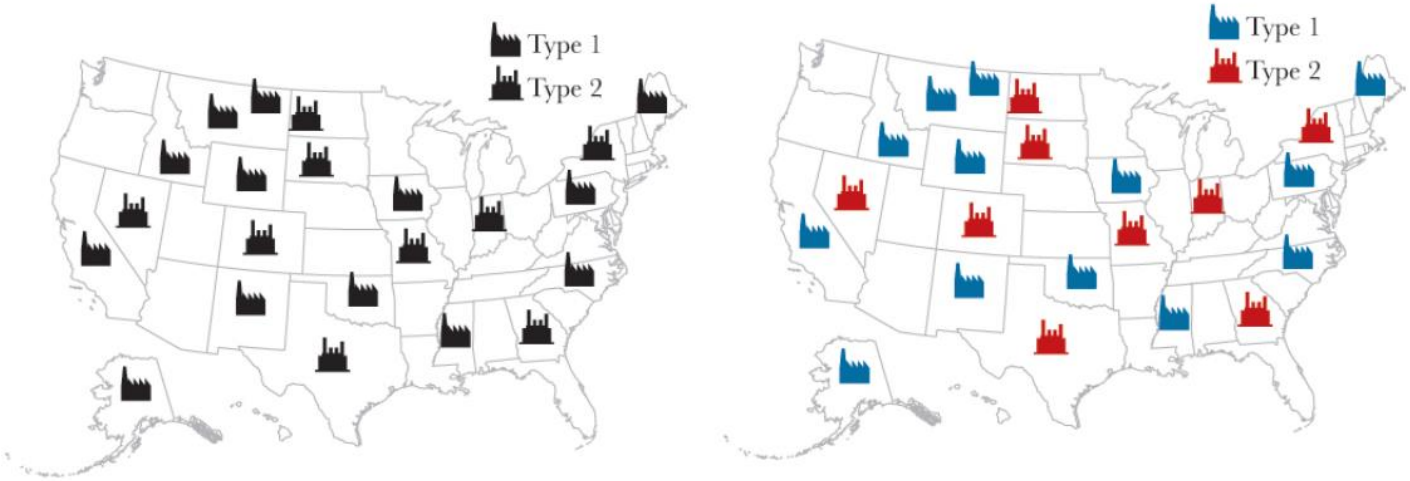


Figure 6.3. On which of these maps is it easier to identify the number of factories of each kind?

Cairo (2012)

Leveraging pre-attentive processing in text

Aim

Become familiar with analysing data visualisations according to data visualisation guidelines.

Purpose

The ability to analyse existing visualisations and provide suggestions for improvement is a valuable skill. It will help you develop an understanding of the principles of good design.

In this assessment you will critique the visualisations presented in a dashboard designed to assist users explore the Australian Safety Enforcement data set. The dashboard divides the data set up into different enforcement types (e.g. drugs, speeding, mobile phone use etc).

Choose ONE of the following enforcement dashboards for your critique:

Mobile phone

Speeding

Non-wearing seatbelts

Later in semester you will be redesigning this dashboard to accomodate updated data collection protocols for the Project. This assessment will help you prepare for the Project and may be used to assist us form Project teams.

This is a individual assessment worth 10% of your final mark.

See Assignments for marking rubrics and submission details.

Suggested Structure

The following is the suggested structure for your assignment.

Section 1: Introduction

What is this dashboard about, what is it's purpose?

Who is the audience for this dashboard?

What questions might the user have about the data set?

Section 2: Data Processing and Governance

Examine the data set/s used for your selected dashboard.

What is the source (or sources) of the data set and what processing has been done on it?

Are there any governance, quality, privacy issues or bias in the data that you can detect? Have they been adequately dealt with or identified on the website?

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Section 3: Visualisation Design

How has the data been mapped to visual elements in the visualisation?

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Draw focus to important part of story

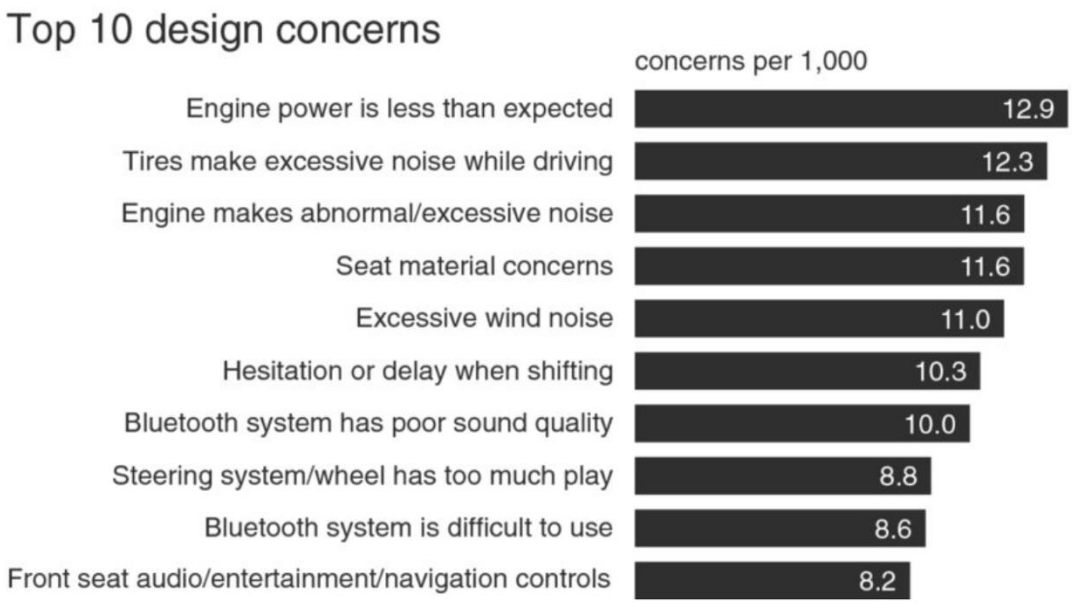


Figure 4.7 Original graph, no preattentive attributes

7 of the top 10 design concerns have 10 or more concerns per 1,000.
Discussion: is this an acceptable default rate?

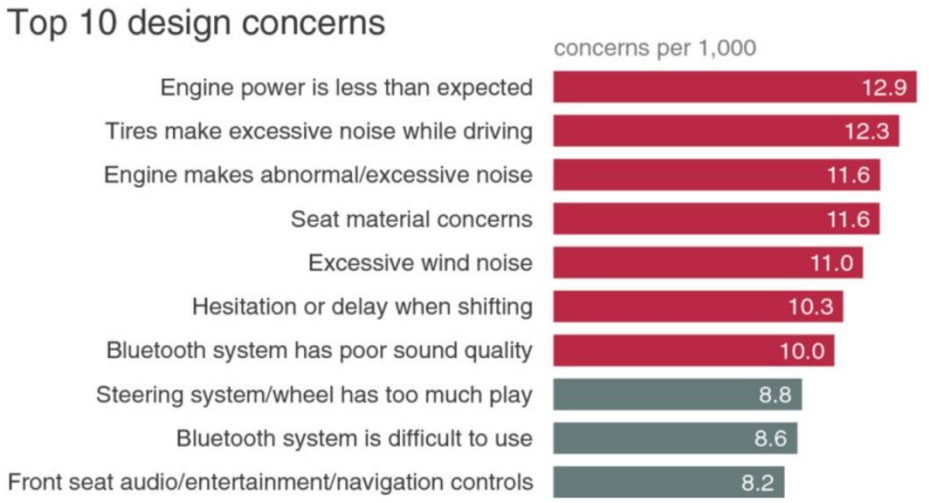


Figure 4.8 Leverage color to draw attention

Knaflic (2015)

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Attention

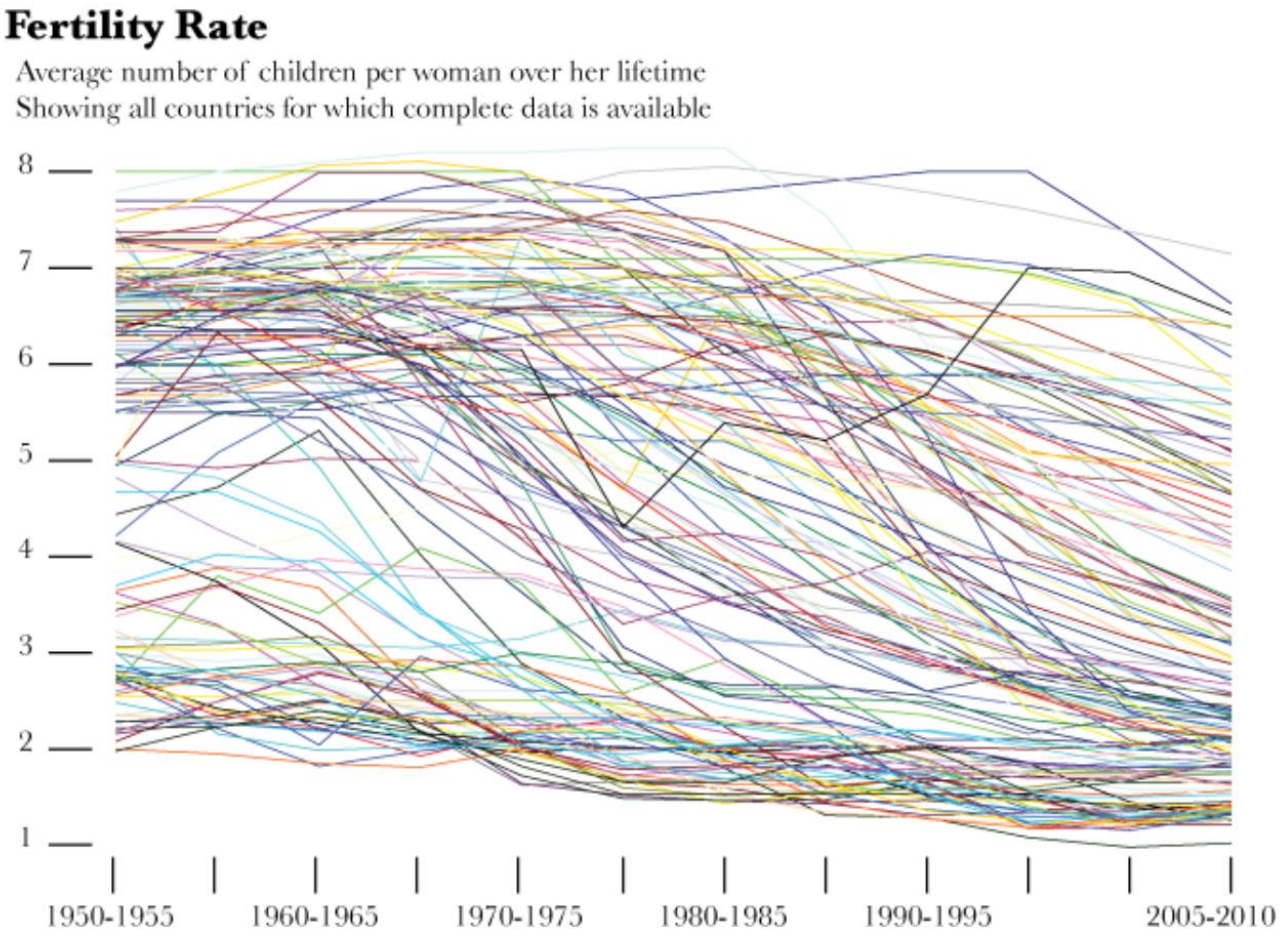


Figure 1.5. Too many lines obscure the message.

Cairo (2012)

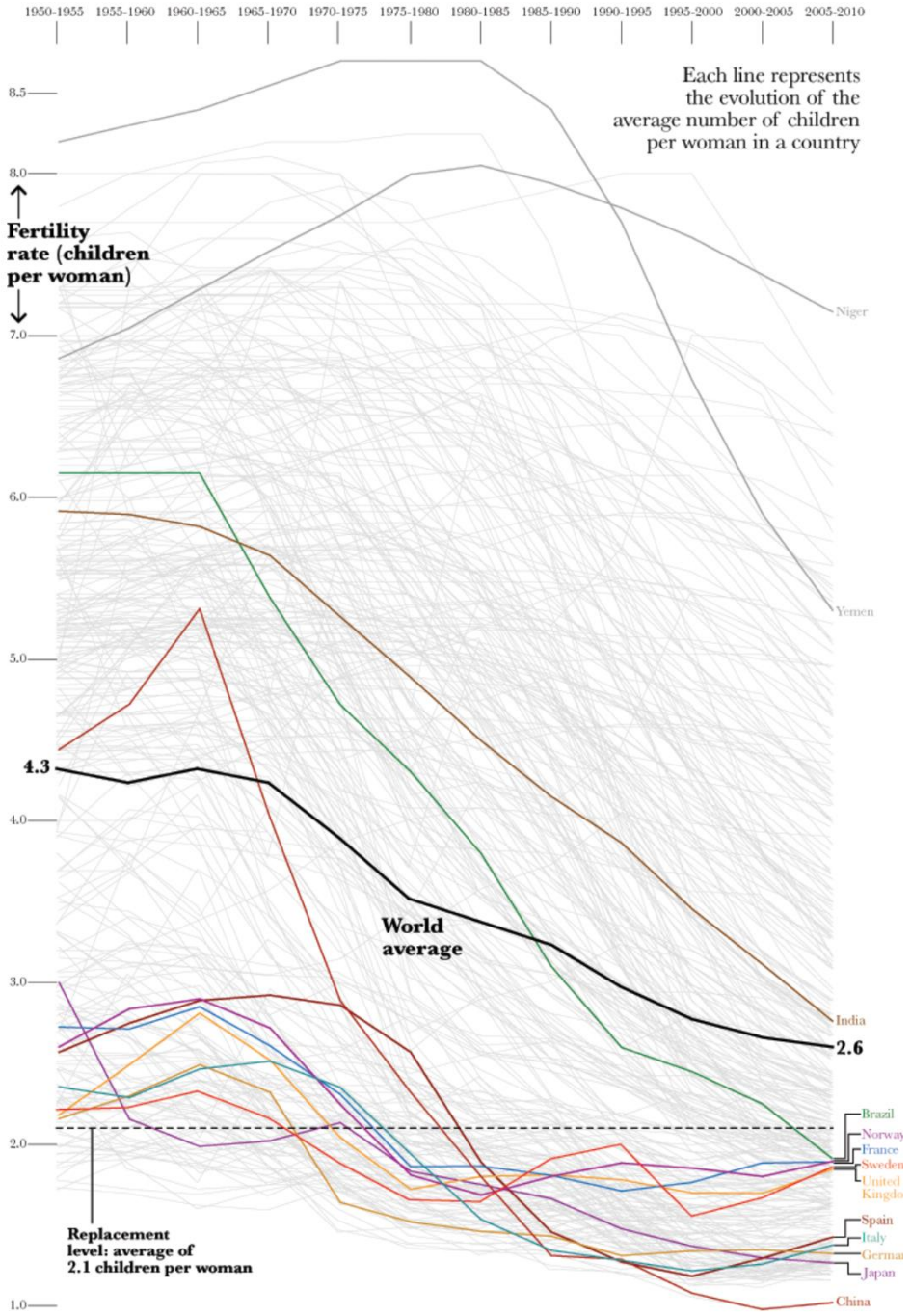


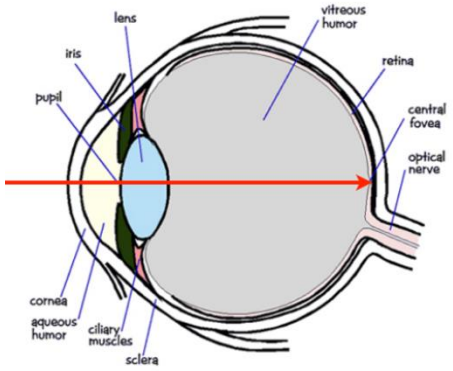
Figure 1.6. Highlighting the relevant, keeping the secondary in the background.

Colour

- Not a big part of normal vision
- Does not help us perceive layout
- Does not show movement or shape
- Helps us pick out elements in our environment (breaks camouflage)
- Can be perceived differently depending on context



www.vischeck.com/examples/



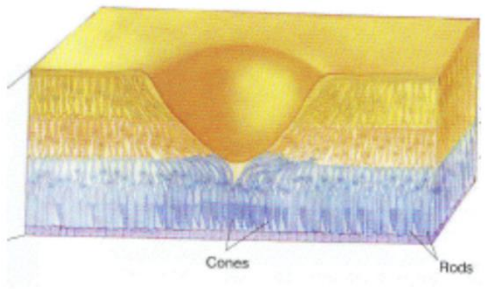




Figure 5.6. On the left, you see what your brain thinks you are seeing. On the right, you see what your eyes are really getting: tons of quick, narrow fixations, such as these views.

Cairo (2012)

Dimensions of colour

Hue

- what most people call ‘colour’

Saturation

- purity of a colour (pale to vivid)

Value (brightness, luminosity)

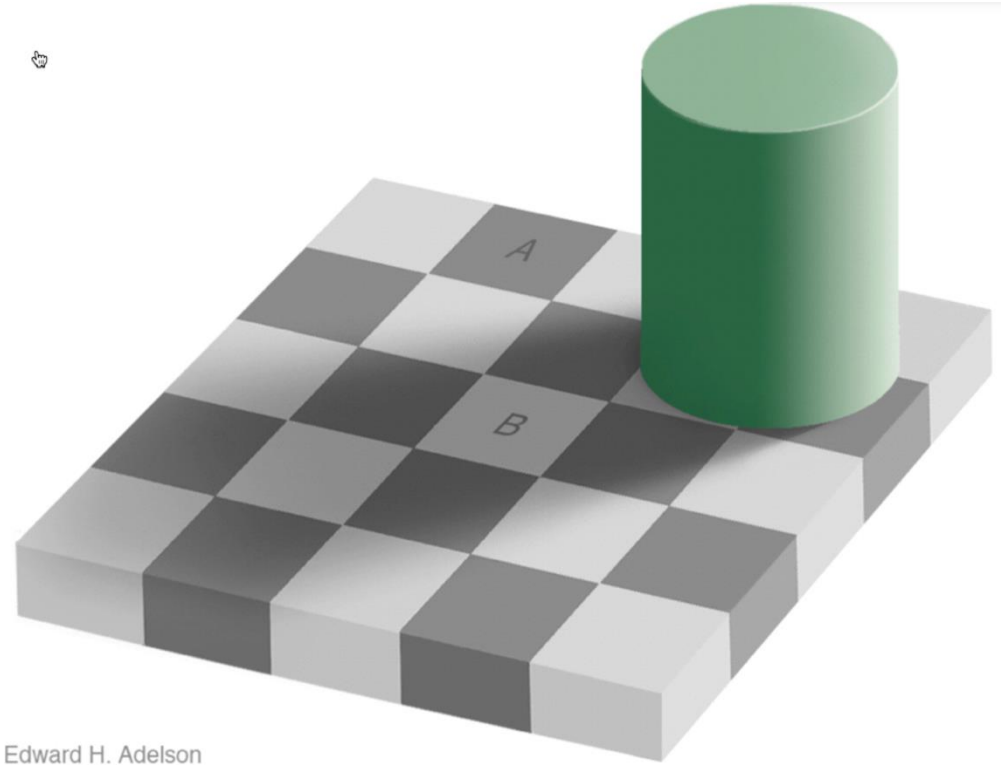
- lightness (or darkness) of a colour (light to dark)



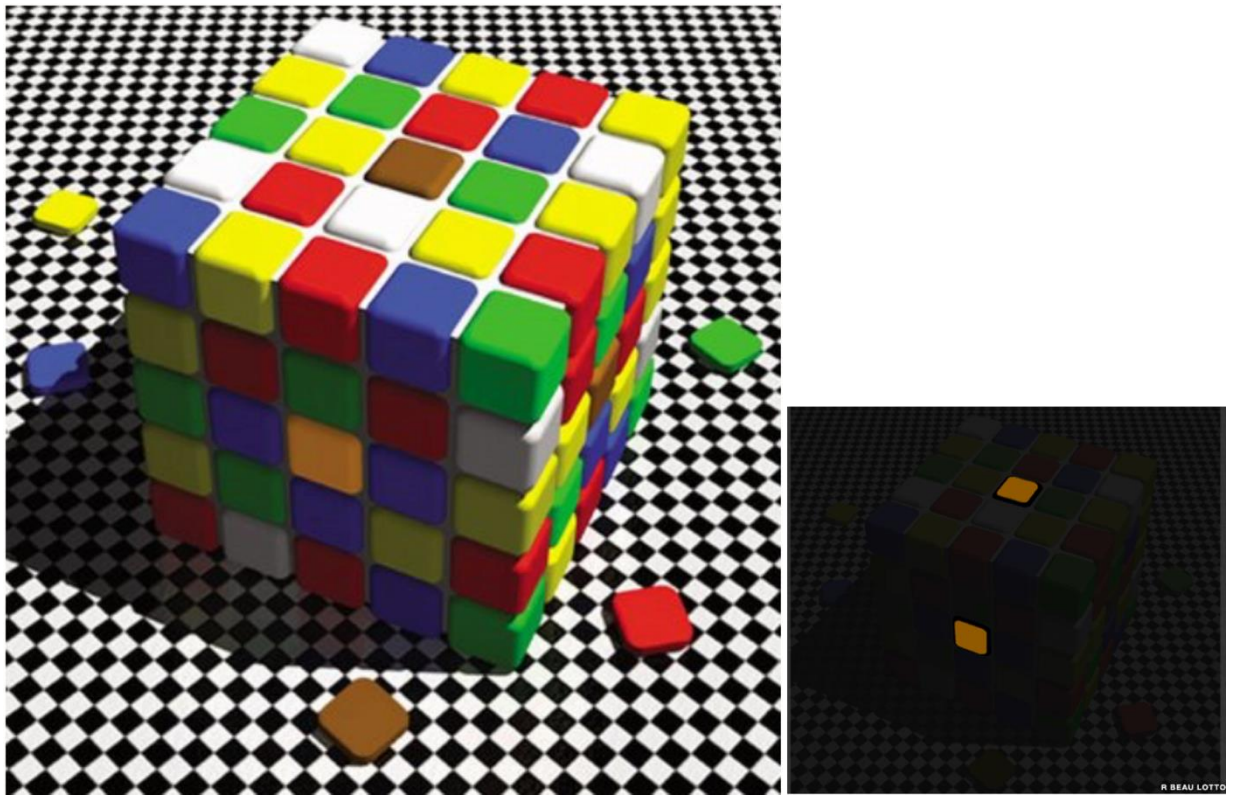




Perception of colour can be relative



Edward H. Adelson



Colour use guidelines

Don’t overuse it (no more than 6-12 different colours)

Harder to see colour on smaller items

Use luminance and saturation for magnitude channels

Use hue for **categories**

Be mindfulness of colour blindness



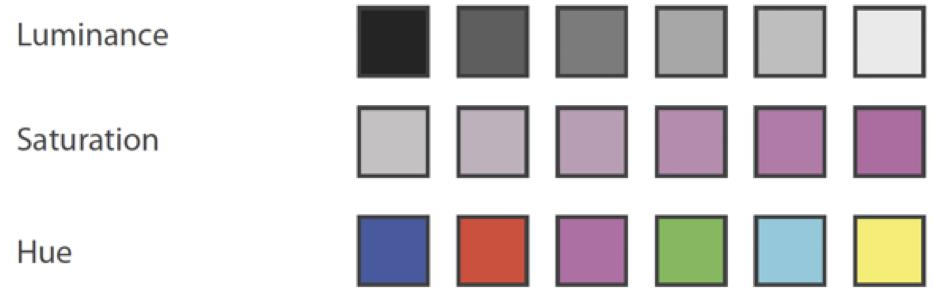
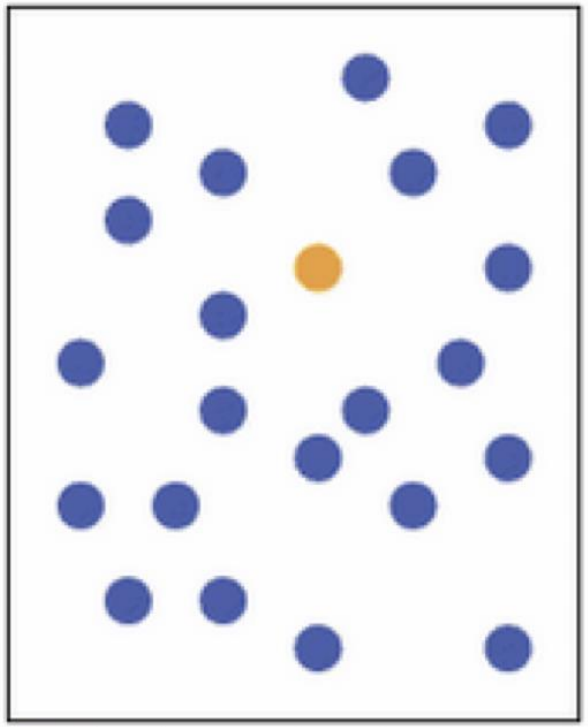
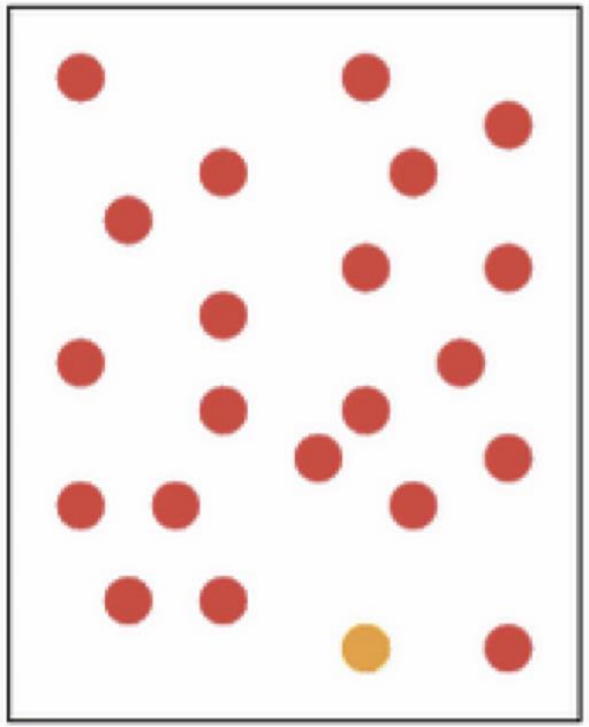


Figure 10.5. The luminance and saturation channels are automatically interpreted as ordered by our perceptual system, but the hue channel is not.

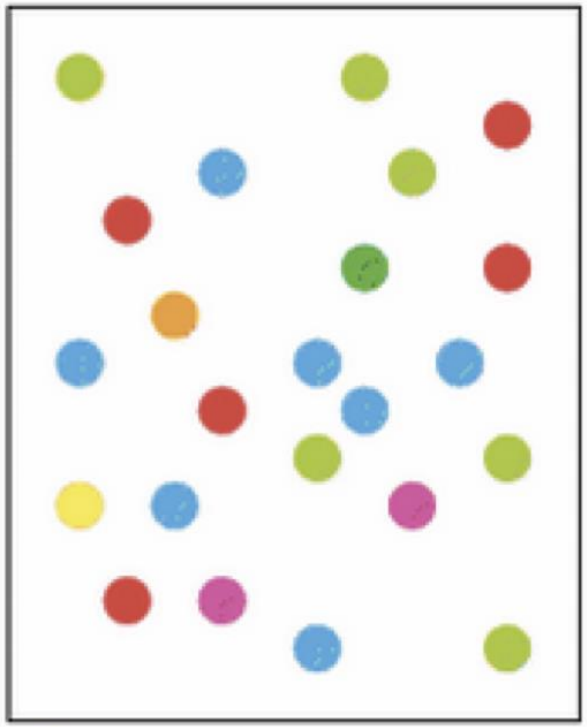
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The larger the chromatic difference between the target symbol and the other symbols, the easier the search.



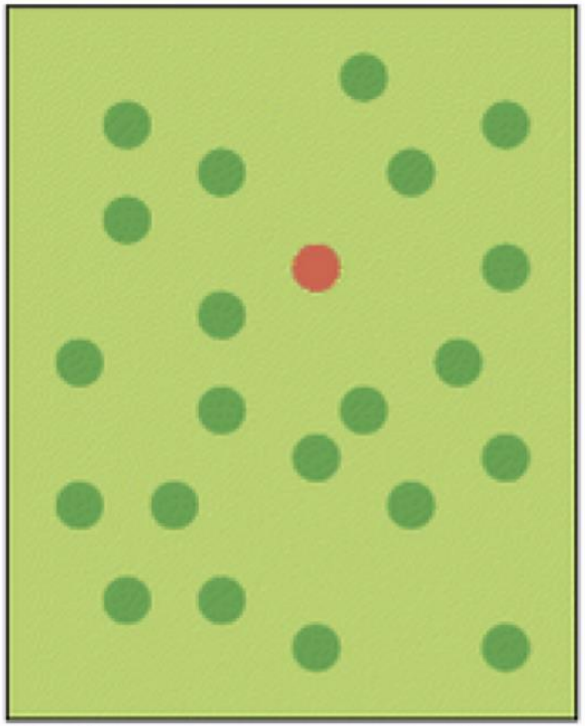
When there is only a small color difference from non-target symbols, the search is difficult.



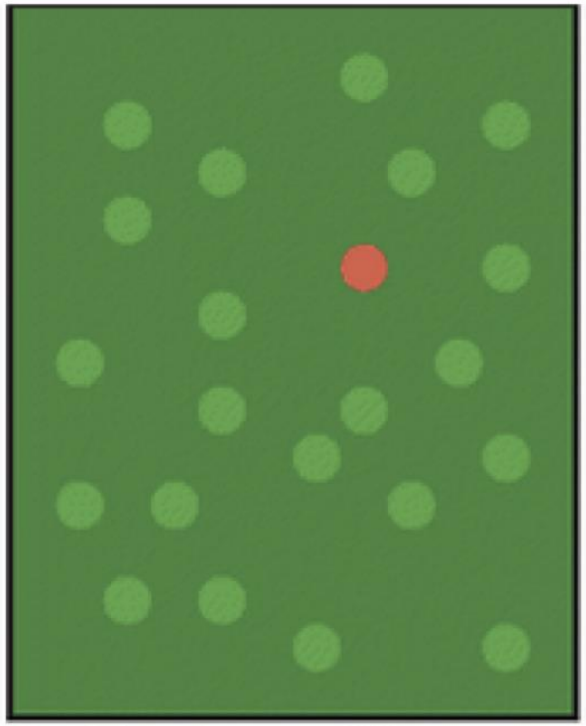
When there are many non-target symbol colors, the search is the most difficult.

Ware (2010)

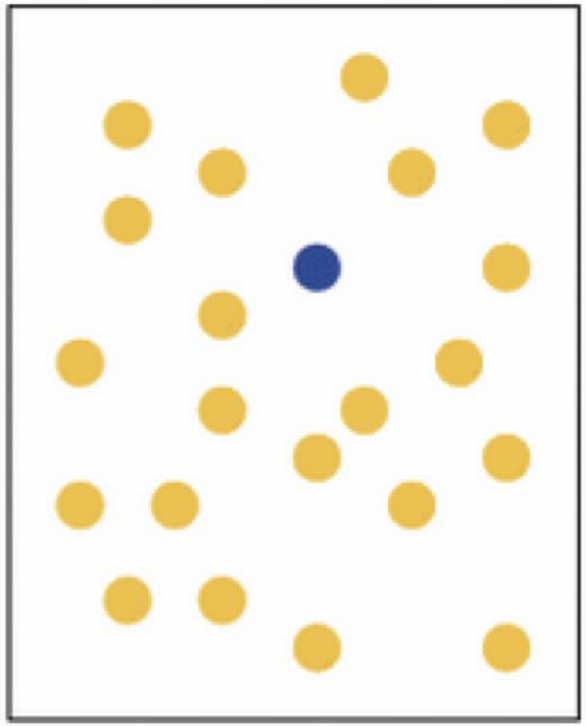
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If non-target symbols are similar to the background, they are easy to exclude from the visual search.



A luminance difference plus a chromatic difference from other symbols and the background leads to the easiest search.

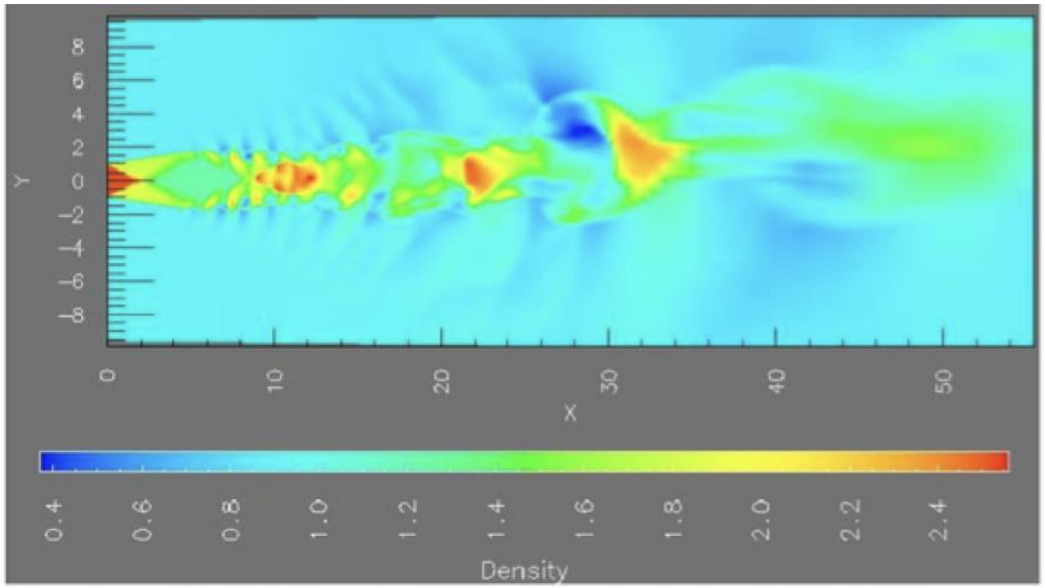


A dark target on a light background with light non-target symbols can be as effective as the reverse.

Ware (2010)

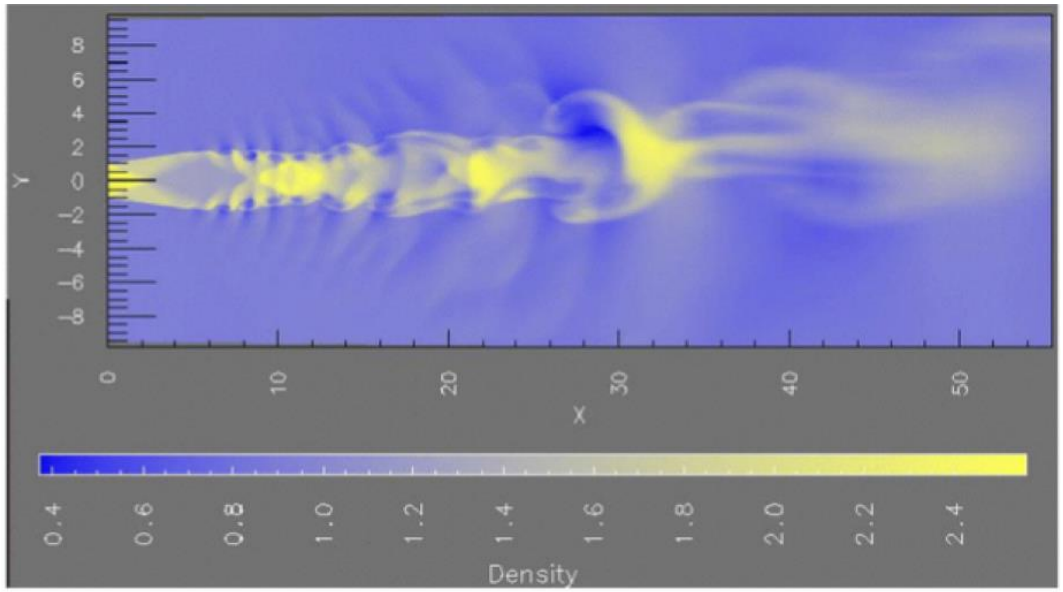
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Divergent Colour Scales



[Rogowitz and Treinish, Why Should Engineers and Scientists Be Worried About Color? http://www.research.ibm.com/people/l/lloydt/color/color.HTM]

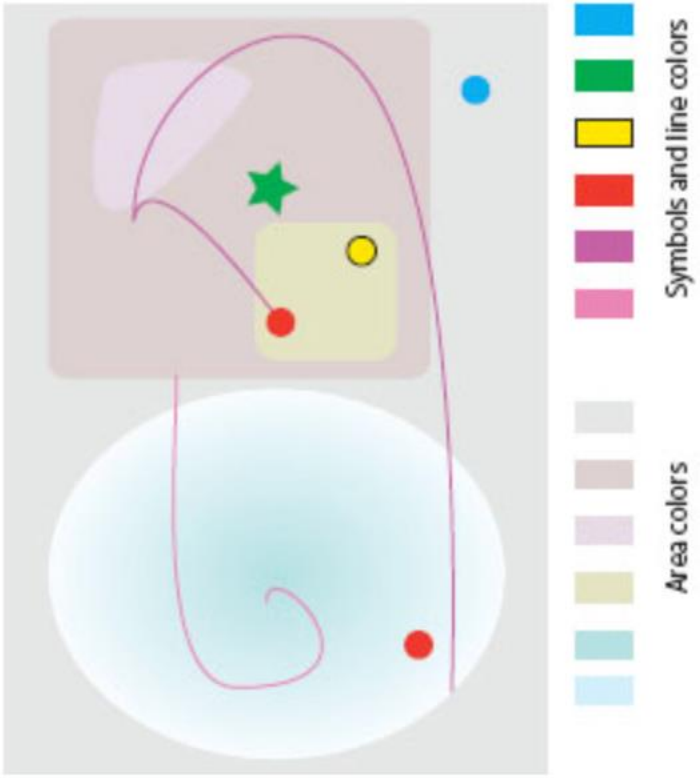
Hue - Rainbow map
difficult to see all features of data



[Rogowitz and Treinish, How NOT to Lie with Visualization, www.research.ibm.com/dx/proceedings/pravda/truevis.htm]

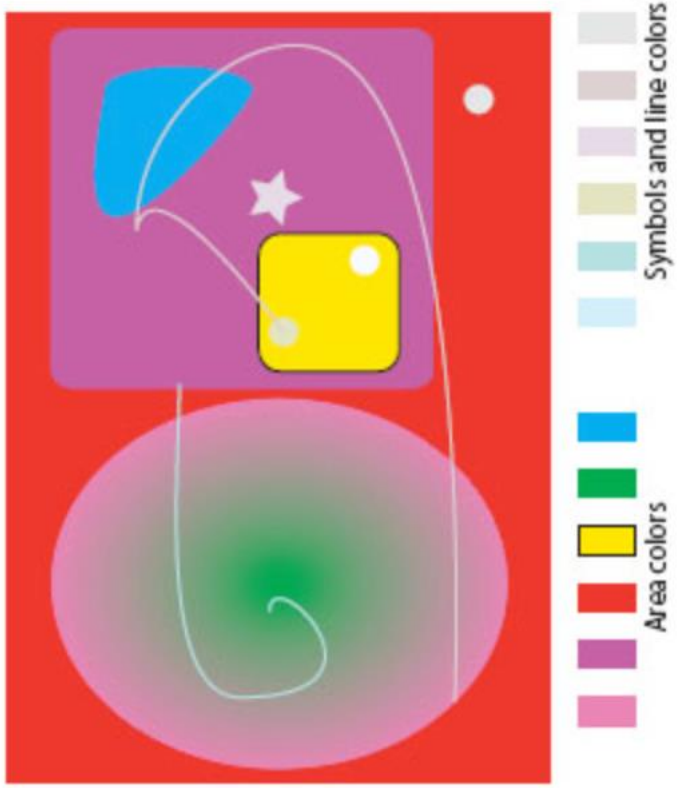
Saturation - Divergent scale (blue/yellow)
easier to see features of data

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"GOOD. This example uses high saturation (strong colors) to code small areas such as symbols and lines. Larger background areas are all light and low saturation.
A black contrasting border is used to separate the yellow circle from Its background because both have similar luminance."

Ware (2010)



"BAD. This example shows what happens when the colors used for small areas and large areas are switched. The color codes used for the symbols and lines are difficult to discriminate because of contrast effects."

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Figure 6.11. Don't overuse background boxes. Doing so will make your graphic look cluttered. If you need to differentiate between different sections, use white space.

Cairo (2012)

Too many differences are difficult to follow

Also… not appropriate to use hue for magnitude

Country Level Sales Rank Top 5 Drugs

Rainbow distribution in color indicates sales rank in given country from #1 (red) to #10 or higher (dark purple)

Country	A	B	C	D	E
AUS	1	2	3	6	7
BRA	1	3	4	5	6
CAN	2	3	6	12	8
CHI	1	2	8	4	7
FRA	3	2	4	8	10
GER	3	1	6	5	4
IND	4	1	8	10	5
ITA	2	4	10	9	8
MEX	1	5	4	6	3
RUS	4	3	7	9	12
SPA	2	3	4	5	11
TUR	7	2	3	4	8
UK	1	2	3	6	7
US	1	2	4	3	5

Figure 4.15 Use color sparingly

Top 5 drugs: country-level sales rank

RANK	1	2	3	4	5+

COUNTRY | DRUG	A	B	C	D	E
Australia	1	2	3	6	7
Brazil	1	3	4	5	6
Canada	2	3	6	12	8
China	1	2	8	4	7
France	3	2	4	8	10
Germany	3	1	6	5	4
India	4	1	8	10	5
Italy	2	4	10	9	8
Mexico	1	5	4	6	3
Russia	4	3	7	9	12
Spain	2	3	4	5	11
Turkey	7	2	3	4	8
United Kingdom	1	2	3	6	7
United States	1	2	4	3	5

Knaflic (2015)

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Using brand colours… or not

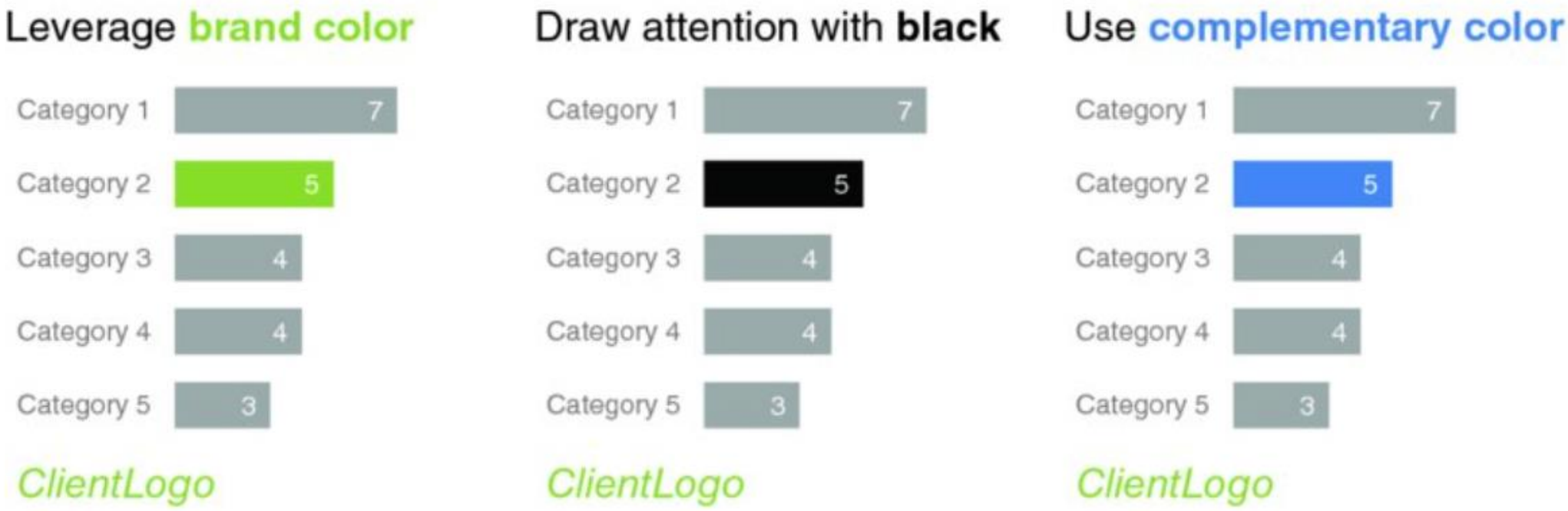
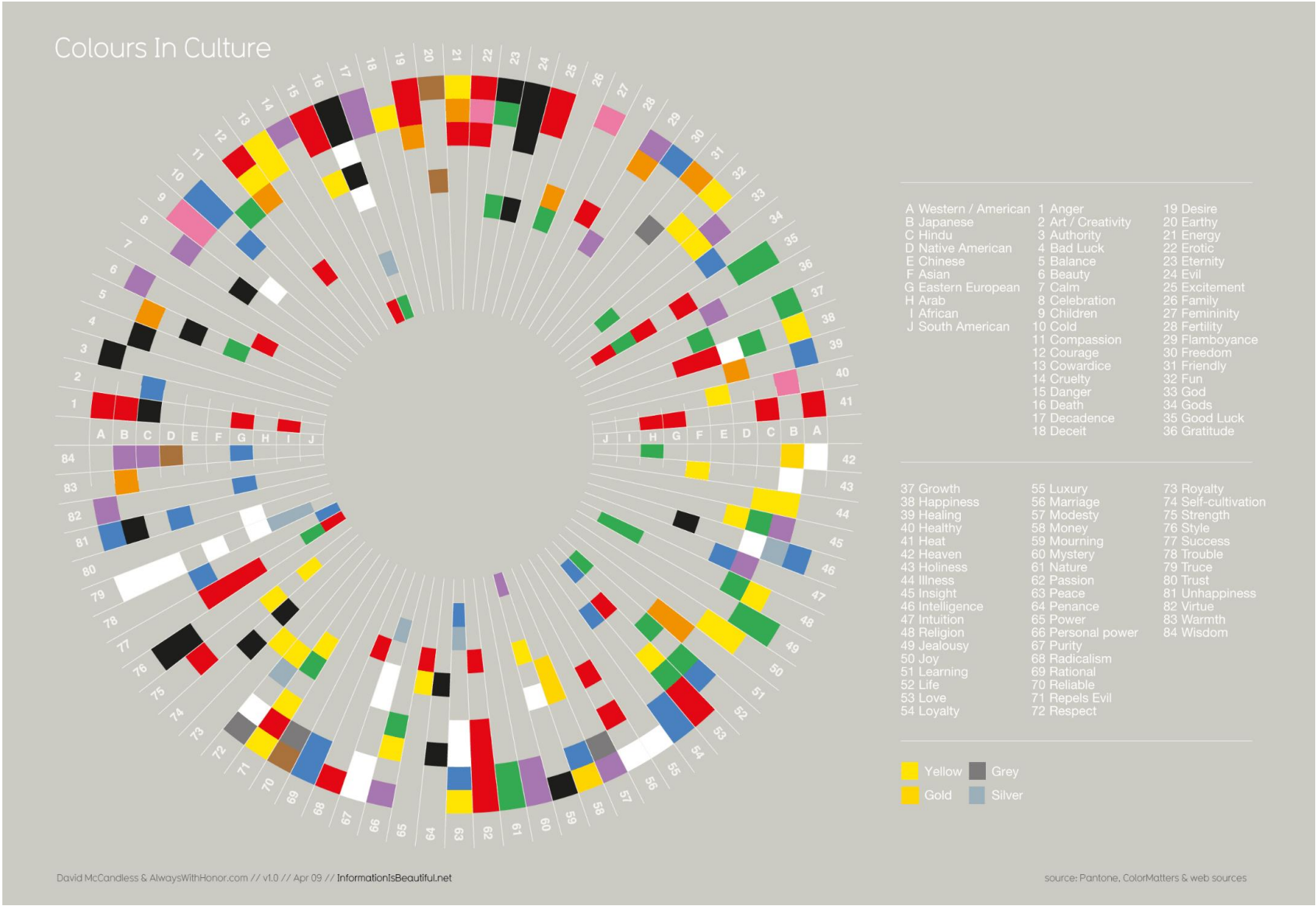


Figure 4.16 Color options with brand color

Knaflic (2015)

Colour Association

Different colours have different associations in different cultures



https://informationisbeautiful.net/visualizations/colours-in-cultures/https://informationisbeautiful.net/visualizations/colours-in-cultures/

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Colour blindness

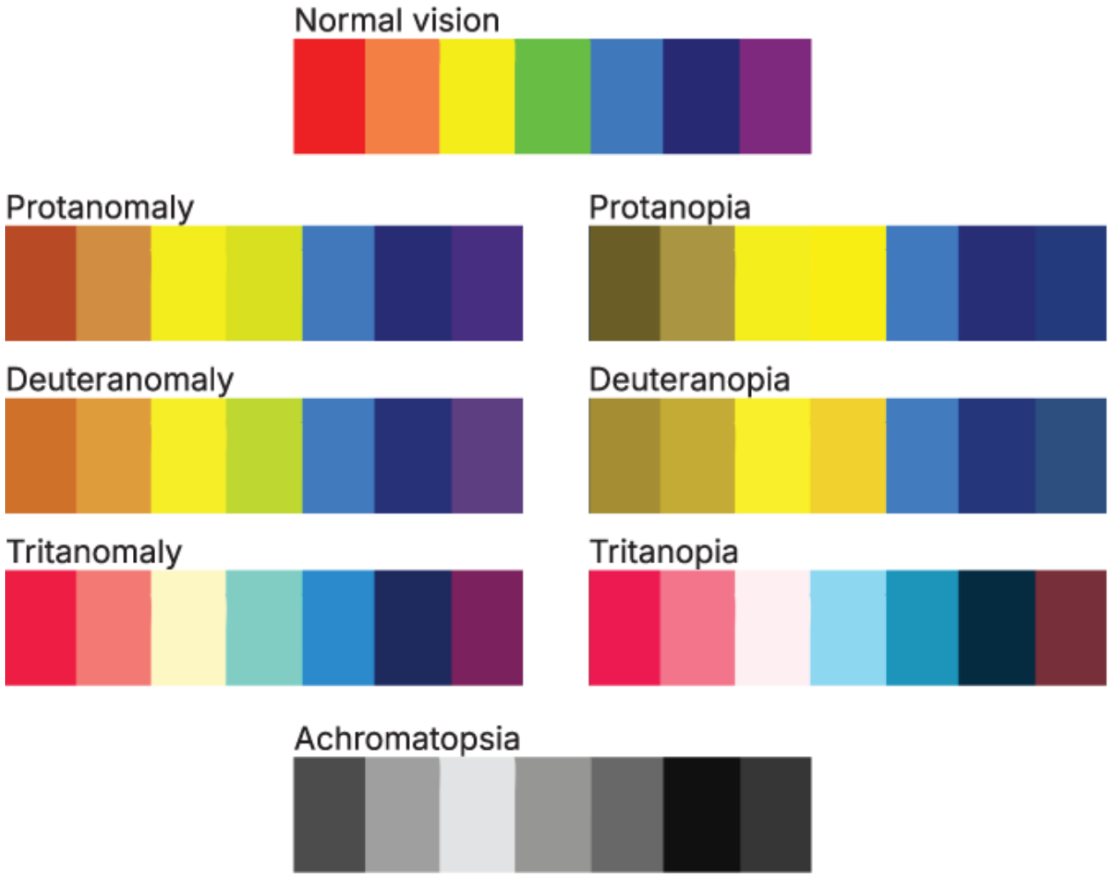
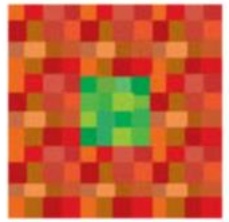
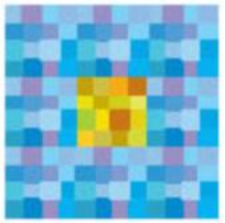


Figure 4.9 An approximate demonstration of what each type of color blindness looks like

Red green colour blind



Can't see green square

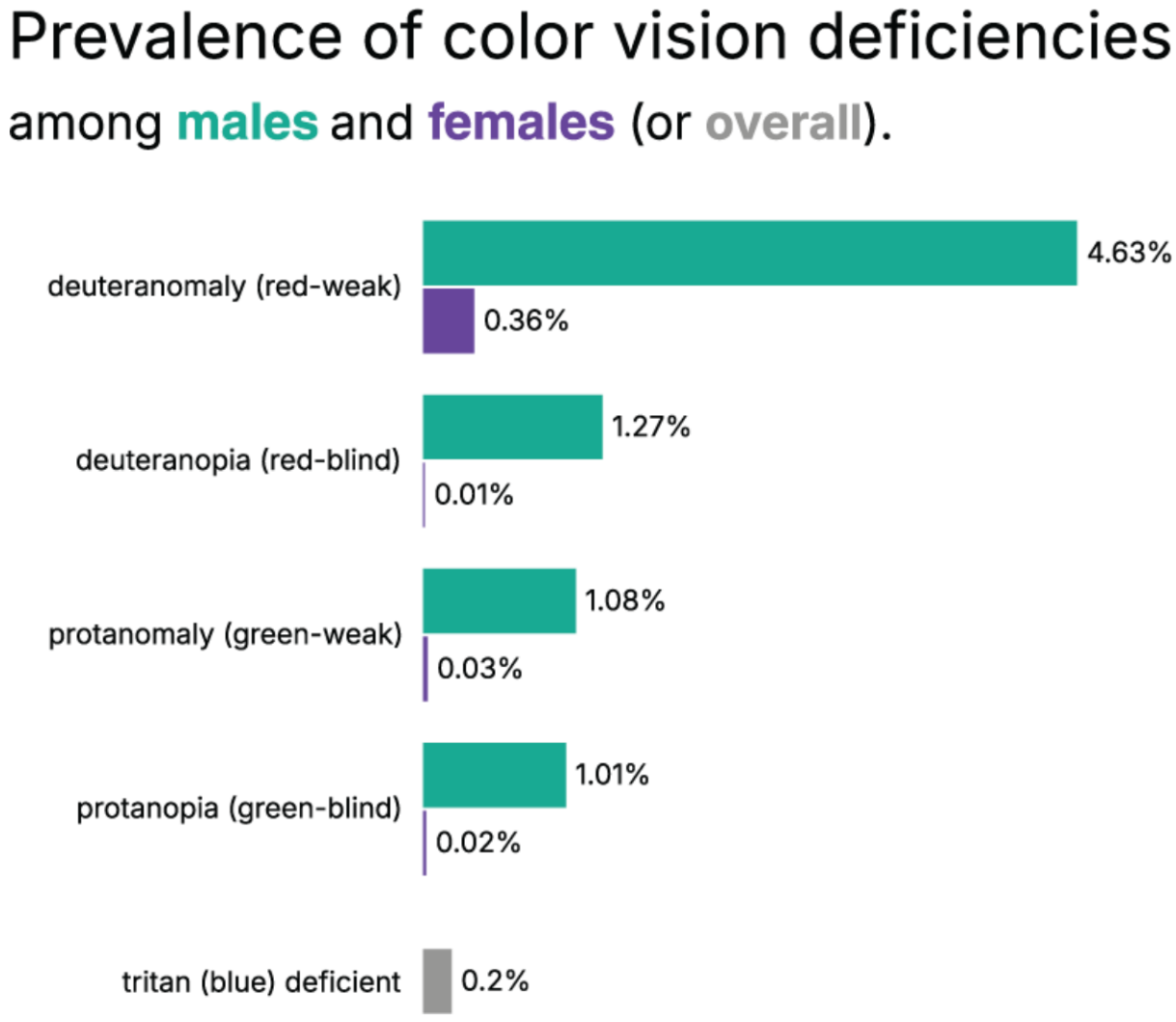


Can see yellow square

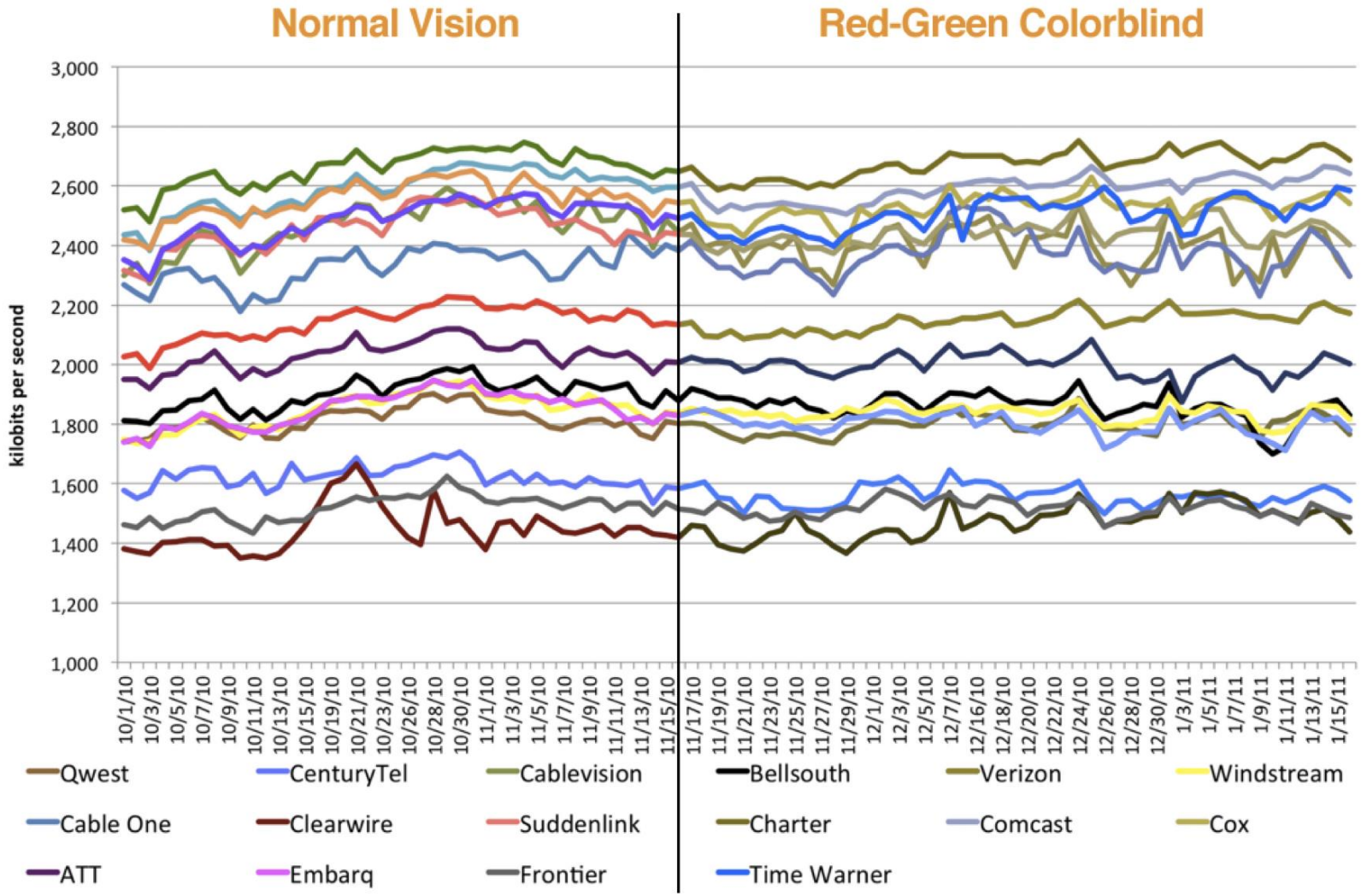
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Colour Blindness

Strong gender differences in rates of colour blindness



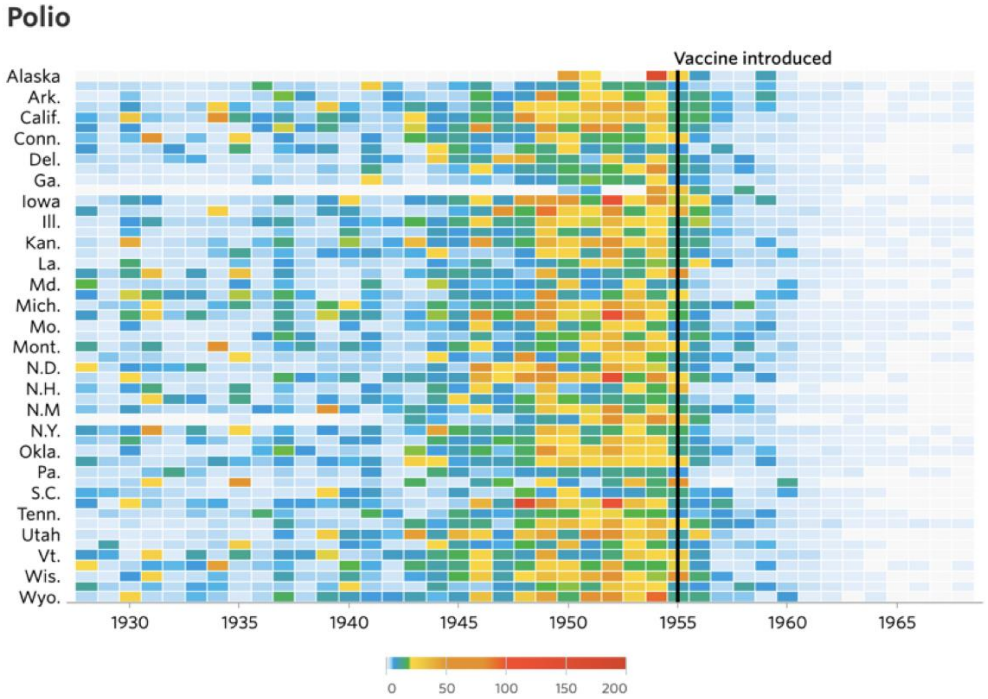
Abbott (2023) Everyday Data Visualisation



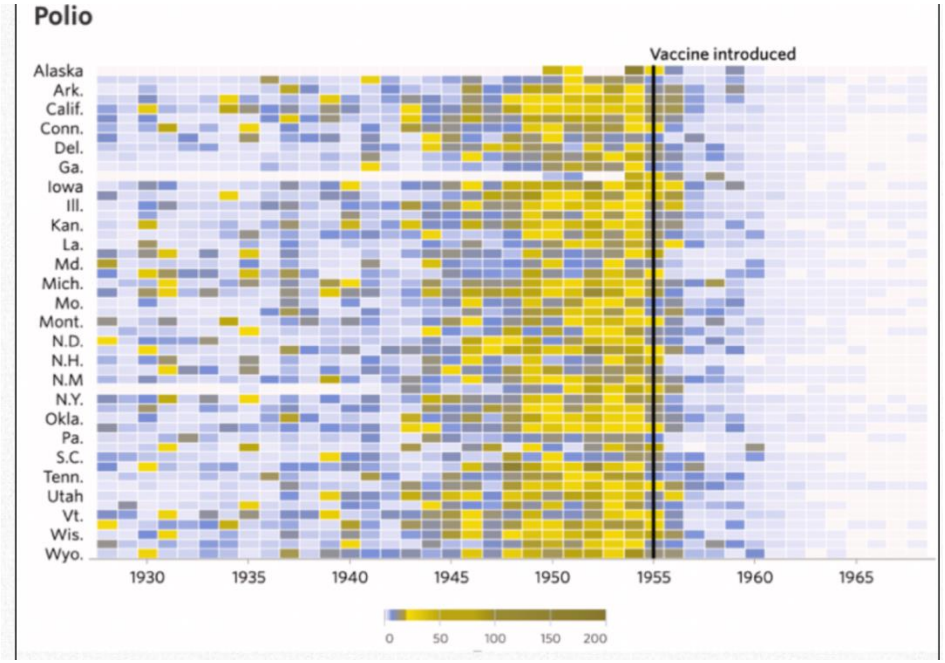
https://waxy.org/2011/01/colorblind_leading_the_blind/

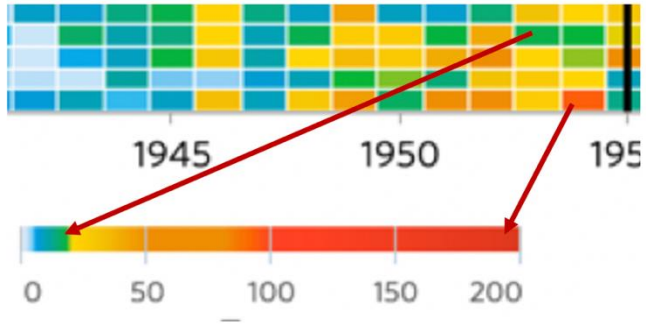
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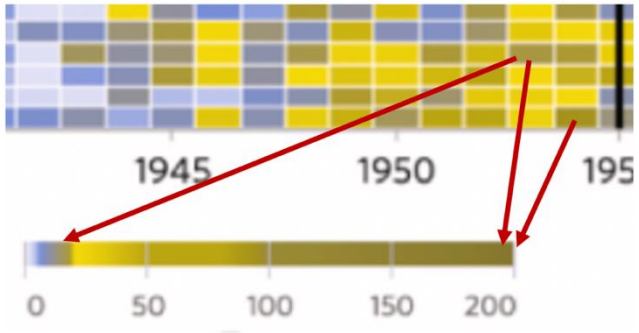
Normal vision

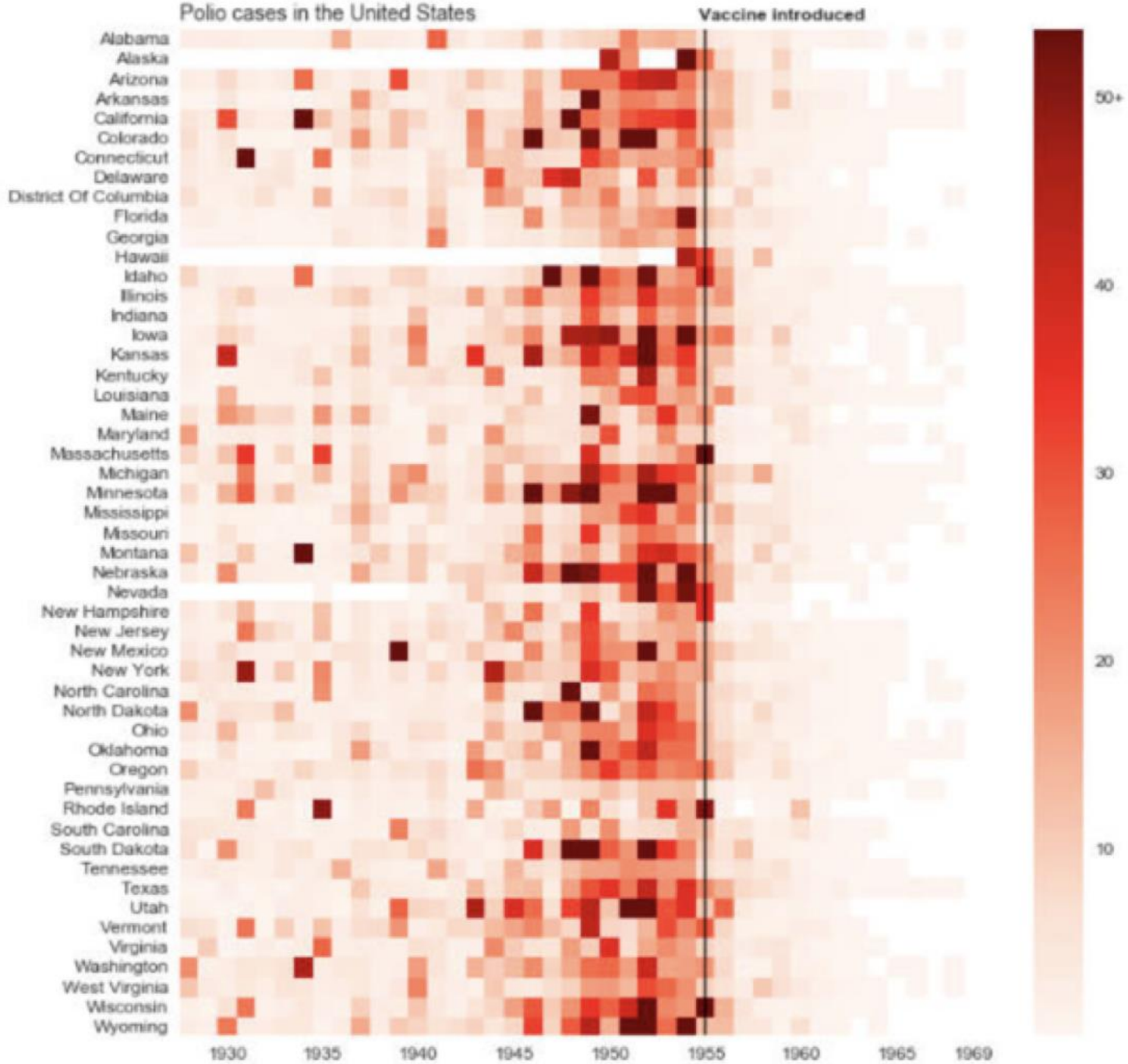


Red Blind - Protanopia









https://www.kdnuggets.com/2016/03/4-lessons-brilliant-data-visualization.html

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Accessibility Guidelines (WCAG)

“Goal

- Color is not the only way of distinguishing information.

What to do

- Use information in addition to color, such as shape or text, to convey meaning.

Why it's important

- Not everyone sees colors or sees them the same way. “

“If content is conveyed through the use of colors that differ not only in their hue, but that also have a significant difference in lightness, then this counts as an additional visual distinction, as long as the difference in relative luminance between the colors leads to a contrast ratio of 3:1 or greater. “

“Color is not used as the only visual means of conveying information, indicating an action, prompting a response, or distinguishing a visual element. “

https://www.w3.org/WAI/WCAG22/Understanding/use-of-color

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